The PQC Migration Prize — the Working Repository

What has been built for the evaluation committee, the metric register, the neutral testbed, and the record — before the call opens

Blockchain Governance Initiative Network

BGIN – Mitchell Travers · IKP Working Group, convener

github.com/bgin-global/pqc-agility-competition | draft v0, Phase A | BGIN Block #15, 15–16 October 2026

Companion to the IKP PQC Migration sessions · everything shown is public and unratified: nothing is final until the committee votes and the host publishes the call

What the Repository Is — and Is Not

The BGIN coordination repository for the *PQC Migration Prize Competition*: a multi-year (2026–2028), internationally open prize on post-quantum migration for public blockchains, hosted by the Japanese Government (METI/NEDO). The repository is **the record**: decisions, drafts, source material, testbed specifications.

It is

- Where the committee is formed and the outcomes are ratified
- Where the metric register, rubric and scheme rulings are drafted by pull request
- Where the testbed harness, result schema and calibration tracks are specified
- Where every commit is signed and signed-off, so the history verifies offline

It is not

- The prize itself — Japan hosts, funds, publishes the call and awards
- A new-algorithm contest — NIST outputs are the reference (gate G1)
- Chain endorsement or a vendor scorecard — BGIN is ecosystem-neutral
- Final — everything is draft v0 until ratified at a workshop

🔗 **Takeaway:** Discussion on the BGIN Discourse; the repository carries the detail.

The Path and the Roles

Competition / evaluation → BGIN Standard → ISO/TC 307 (BGIN Category A liaison) → Japan crypto-asset security criteria

Lane	Role
Japanese Government (METI/NEDO)	Host and organizer: publishes the call, sets rules, funds and awards the prize; later channels results into Japan's crypto-asset security criteria
BGIN	International coordination, neutral evaluation testbed, multi-stakeholder dialogue; produces candidate BGIN Standards with a path to ISO/TC 307
NIST (not a competition lane)	Algorithm and parameter authority for the PQC primitives in scope; measurement-methodology cooperation sought where appropriate

🔗 **Takeaway:** The hard problem on public chains is deployment, evaluation, testbeds, and migration governance — not selecting new primitives.

Repository Map — Five Jobs, One Tree

1 Committee	2 Outcomes	3 Metrics + rubric	4 Testbed	5 Workshops
GOVERNANCE.md COMMITTEE.md evaluation/ COI_POLICY.md decisions/	OUTCOMES.md O1–O6 + non-outcomes resources/ (briefing, digest, provenance)	evaluation/ METRICS.md M-01..M-22 evaluation/ RUBRIC.md gates G1–G4	testbed/ README, SCHEMES, HARNESS schema/ result.schema.json tracks/ (T1) harness/ (instance)	workshops/ ROADMAP.md W1–W6 workshops/ TEMPLATE.md outreach/

Integrity and community layer DCO.md · docs/SIGNING.md · docs/SECURITY_MODEL.md · docs/CONFIDENTIALITY.md · docs/AUTORESEARCH.md · .github/rulesets · .githooks + scripts · CONTRIBUTING, SUPPORT, MAINTAINERS, CHANGELOG, CITATION.cff · Apache-2.0 code, CC BY 4.0 documents

🔗 **Takeaway:** The README numbers the five jobs; every directory answers to one of them. Draft v0 throughout.

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Job 1

Form the Evaluation Committee

Modelled on a NIST-style process: open criteria, published rationale, reproducible results, mandatory conflict-of-interest handling.

Governance — Phases and Composition

Phase	Window	Output
A — Criteria	Sep 2026 → Jan 2027	Outcomes ratified; metrics and rubric drafted; scheme rulings; testbed harness; COI policy; call-text inputs to the host around SSR (Dec 2026)
A2 — Pilot	Jan → Sep 2027	Validated harness; node roster; rubric frozen before the application deadline
B — Evaluation	Nov 2027 → mid 2028	Theoretical review track and testbed performance track; evidence set citable in standards
C — Standards	Oct 2028 →	BGIN Standard candidate → ISO/TC 307 → Japan criteria

Composition (proposed, 12–18 voting members): 2 conveners · 3–5 academia / security review · 2–3 Bitcoin developers · 2–3 Ethereum developers · 2 wallet and infrastructure · 2 measurement / testbed · 1–2 governance / policy · technical advisors without votes · observers: host, NIST liaison, ISO/TC 307 liaison.

🔗 **Takeaway:** Geographic and ecosystem balance is a seating criterion. Phase A membership does not oblige Phase B.

How Decisions Are Made — and Kept Neutral

Decision process (GOVERNANCE §5)

1. Discuss on Discourse — every proposal starts as a thread
2. Record by pull request, citing the thread and the workshop
3. Rough consensus decides drafts; objections recorded
4. Formal votes for: ratifying outcomes, freezing the rubric, scheme scope rulings, amending governance or COI. Simple majority, quorum of half; conveners vote only to break ties
5. Every vote and scope ruling gets an ADR in decisions/

Neutrality (COI_POLICY, adopt at W3)

- Mandatory COI disclosure for evaluators and node operators, published in the roster
- Submitters and close affiliates do not judge their round
- Node operators running an artifact under evaluation are recused
- Host funding and BGIN coordination must not compromise technical neutrality
- Measurement seat: node hosts are not solvers on the tracks they measure

🔗 **Takeaway:** Neutrality is what makes the outputs usable as standards inputs rather than a private scorecard.

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Job 2

Identify the Core Outcomes

Six outcomes, each with a definition of done, the workshop that ratifies it, and an owner seat. Seeded from the METI/NEDO community briefing and the IKP-WG discussions.

Outcomes O1–O6 (draft v0, for ratification at W1)

	Outcome	Done when	Ratified
O1	Evaluation framework in standards-ready vocabulary	METRICS and RUBRIC frozen by vote before the application deadline; every metric has a unit, method, mapping	W2; frozen A2
O2	Neutral, reproducible testbed evidence	Harness and schema adopted; $\geq$ 3 pilot nodes in 2 regions reproduce a reference run within tolerance	W3; pilot A2
O3	Crypto-agility evidence, not only readiness	Register carries switch, rollback and staged-migration metrics the testbed can execute	W2
O4	Migration-governance evidence	Governance-fit dimension with a qualitative method; open positions captured in a record	W3–W4
O5	Call-text inputs delivered to the host	Versioned package tagged; receipt acknowledged by METI/NEDO	W4
O6	Standards path	BGIN Standard outline v0; maintainer group named for Phase C	W6

Non-outcomes: new primitives or parameter sets · chain endorsement or vendor scorecards · a Q-day date · prize adjudication.

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Job 3

Draft the Evaluation Metrics and Rubric

A 22-row register with units, methods and target mappings; a rubric skeleton with four gates. Written to travel: competition to BGIN Standard to ISO/TC 307 to Japan criteria.

The Metric Register — M-01 to M-22

Dual track: **(a)** theoretical security review, **(b)** implementation and performance on the testbed. Each row names the mapping it needs: BTC-tx, ETH-user, ETH-cons, L2.

Group	Rows
Sizes and single-op cost (b)	M-01 signature size · M-02 public key size · M-03 sign time · M-04 verify time · M-05 batch / aggregate throughput · M-06 aggregate size
Chain-level overhead (b)	M-07 per-block overhead at a reference tx mix · M-08 mempool / gossip propagation · M-09 block validation time · M-10 on-chain verification gas · M-11 state growth · M-16 L2 / ZK verification cost · M-17 decentralization cost
Agility (a+b)	M-12 migration throughput · M-13 rollback cost · M-14 agility switch cost · M-21 legacy verification continuity
Review and fit (a)	M-15 wallet / UX impact · M-19 security review score · M-20 governance fit · M-22 PQC-readiness claim substantiation
Gate metric (b)	M-18 reproducibility — nodes and regions reproducing the reference run within tolerance

🔗 **Takeaway:** Performance rows are scored relative to the classical baseline on the same node, never as absolute numbers. Proposed directions per row are on the register for W2.

The Rubric Skeleton — Gates, Tracks, Dimensions

Gates (must pass before scoring)

- **G1 Primitive scope** — NIST-standardized or selected, or a committee-ruled hybrid / stateful hash-based scheme
- **G2 Target mapping** — evidence maps to BTC-tx, ETH-user, ETH-cons or L2
- **G3 Reproducibility** — k nodes, ≥ 2 regions, within tolerance (M-18)
- **G4 Disclosure** — submitter COI disclosed; evaluators recused as required

Dimensions (weights TBD at W2–W4)

- (a) assumptions, parameters, composition · threat model · readiness substantiation
- (b) verifier cost and latency · block / mempool overhead · on-chain and L2 cost · decentralization cost
- (a+b) crypto-agility: switch, staged migration, rollback, legacy continuity
- (a) compatibility and wallet / UX · governance fit

Two evaluators per dimension; a 5-band scale with written justification; rationale published with the score.

🔗 **Takeaway:** Rationale published with the score is what makes a result citable. Weights, k, tolerance, reference CPU class and block mix are decision records.

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Job 4

Specify the Crypto-Agility Testbed

Servers exercising pre-existing signature schemes in Bitcoin- and Ethereum-mappable configurations. One interface, one manifest, one result schema, one reproduction protocol: that is what turns node runs into evidence.

BSafe.network — the Neutral Testbed

- Geographically distributed, neutral nodes: on the order of 10–15, multi-region, multi-month runs
- Hosts: accredited universities, non-profits, neutral R&D labs; contracting later under the host process
- Role separation: node operators are not judges
- Funding: initial nodes by the host; additional by contributing organizations; disclosed per node
- Node requirements by decision record: reference hardware class, pinned OS image and container digests, NTP and region label, operator signing key, node peering for M-08; no mainnet dependence

Phase	Window	Nodes
T0 Spec	Sep–Nov 2026	none; a hosted development runner for pre-flight
T1 Pilot	Jan–Sep 2027	3–5 volunteer
T2 Evaluation	Nov 2027 – mid 2028	10–15 contracted

Node roster: empty; offers via the *Node host offer* issue template. The call opens at W3.

🔗 **Takeaway:** What runs on a node: the reference harness over the scheme register, scenario runs for the chain-level metrics, and in Phase B the submitted artifacts under the same manifest.

The Scheme Register — S, B and C Rows

Row	Scheme	Reference	Status / note
S-01	ML-DSA (44 / 65 / 87)	FIPS 204	candidate; the briefing's starting point
S-02	SLH-DSA (128s / 128f / 192s / 256s)	FIPS 205	candidate; conservative, large
S-03	FN-DSA / Falcon (512 / 1024)	FIPS 206 draft	candidate; smallest lattice sigs; float risk
S-04	XMSS / XMSS^MT	SP 800-208, RFC 8391	candidate; stateful; Poseidon2 params need a ruling
S-05	LMS / HSS	SP 800-208, RFC 8554	candidate; stateful; state reuse = key recovery
B-01	secp256k1 ECDSA	Bitcoin legacy / SegWit v0; ETH EOAs	baseline for BTC-tx, ETH-user
B-02	secp256k1 Schnorr	BIP-340, Taproot	baseline for BTC-tx
B-03	BLS12-381	Ethereum consensus	baseline for ETH-cons
C-01	Hybrid classical + PQ	IETF LAMPS direction	ruling; "hybrid then cutover"
C-02	Quantum-resistant output type	P2QRH / BIP-360 direction	ruling; reference only, no endorsement
C-03 / C-04	Account abstraction; BLS to hash-based aggregation	Ethereum	ruling
C-05	Lattice zkVM and client-side proving	September 2026 evidence	ruling

🔗 **Takeaway:** Baselines are measured for the deltas, never as candidates. Rulings are formal votes with records.

The Reference Harness — Interface, Manifest, Reproduction

Common interface

Every backend exposes keygen (deterministic from seed), sign (stateful schemes expose and persist state; the leaf index is recorded), verify, sizes; batch and aggregate optional and reported as unsupported rather than absent.

Run manifest and result

run_id, scheme row, parameter set, implementation (repo, commit, build flags), harness version, node (host, region, hardware fingerprint, image and container digests), fixtures, operator key — hashed before the run; results in schema/result.schema.json.

Reproduction protocol

A reference run is declared; k independent nodes in ≥ 2 regions execute the same manifest; a measurement reproduces when each node's median is within the tolerance band. The count is M-18; it feeds gate G3. Disagreements are published too.

Fixtures: 32-byte sighash-like and tx-hash-like digests plus a variable-length corpus; key sets $n \in \{1, 64, 1024, 10^5\}$; a reference block / tx mix by decision record; a legacy corpus for M-21; and (proposed) banks rather than sequences, with the instance drawn from the submission digest.

🔗 **Takeaway:** Code follows the spec, not the other way round. Implementation starts after W3 adoption.

Decision 0001 — the Open-Benchmark Model for Calibration Tracks

Proposed for W2. The testbed is already a measuring instrument with a reproduction rule; naming a promotion rule makes it a benchmark. Calibration tracks carry no prize.

A track is one directory, one contract

- benchmark.json — editable paths, pinned commands, score file, direction, promotion bar, runner
- RULES.md — what may be edited, the score formula, the anti-overfitting rule
- runner.json — the runner the score is valid on
- fixtures/ — public, with provenance; promoted/ — one directory per promotion

Rules that carry

- Promotion bar = the track's reproduction tolerance; reproduced on *k* nodes *and* better than the incumbent
- Public note $\geq$ 5 KiB naming the exact model and harness used, or "none"
- One lever per submission — the promoted history is read by diffing
- Out of scope: any leaderboard of prize entries; any judgement of an applicant

🔗 **Takeaway:** Whether agents are used is not the committee's to decide — they already produce the field's numbers. Where their output is admissible, and under what attribution, is.

First Tracks — T1 Exists to Be Boring

Track	Metric → scalar	Calibrates
T1 verify-single (drafted)	M-04 median single verify of ML-DSA-65 over B-01, on the reference CPU class; lower is better; edit only backends/ml-dsa/; KATs pass, build pinned, no network	the reference CPU class, the tolerance band, the runner spec — before anything harder depends on them
T2 verify-batch	M-05 at $n \in \{64, 1024\}$; batch equals sequential on the same inputs	whether the register needs a batch row per scheme
T3 legacy-continuity (v1)	M-21 pass / fail + cost on C-01 / C-03 paths	the “PQC-ready chain” continuity clause
T4 agility-switch (v1)	M-14 switch cost; rollback works (M-13)	O3 evidence; whether agility is measurable at all

The draw as the Gap: fixtures are banks of 65,536; a node times a 10,000-index instance drawn from the submission’s own digest and a round secret it does not see. Nothing to tune to.

🔗 **Takeaway:** T1 is small, fast and fully public. Its only job is to make the reference runner real.

The Calibration Harness in the Record — harness/

The instrument is part of the record. harness/upstream/ vendors the agentprivacy dual-agent harness verbatim at a pinned commit (UPSTREAM.json, sha256 per file, known defects and their fixes); harness/instance/ is the pqc-calibration instance: proposer = submitter, prover = testbed node, the Gap = the fixture draw.

- **Runnable as plumbing today:** the pin verifies; the vendored engine passes its own eleven gates; a stub round derives seeds and lands files; every seed and draw re-derives from the saved bytes.
- **Conformance fails by design** on the unmeasured baseline — UNMEASURED until harness v0 exists; CI reports it, enforcement starts once measured.
- **Two defects found on the first stub round, fixed upstream the same day:** a one-byte-per-pick draw (biased for a large bank) and a one-byte seed counter; draw v2 uses 32-bit rejection-sampled picks.
- **What it is not:** the competition's evaluation. Prize entries are measured and attested under the host's rules and never promoted into one another.

🔗 **Takeaway:** Until real nodes are seated, every verdict is a claimed score, never a promotion. The door stays shut until the assay seat runs on a testbed node.

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Job 5 and the record

Run the Workshop Series; Keep the Record Honest

Six workshops ratify jobs 1–4 before the call opens. The integrity and confidentiality layers are what let a neutral claim be checked rather than trusted.

Workshop Roadmap W1–W6, Sep 2026 → Feb 2027

Workshop	Anchor	Outputs / decisions
W1 Kickoff (late Sep)	GDC26 report; CBT session	Repository chartered; OUTCOMES ratified; governance v0; nominations open; conveners named
W2 Committee & metrics (mid Oct)	**Block 15**; EF Quantum Retreat; CSS / BWS	Committee seated; chairs elected; register v0 → v1; scheme scope rulings; open questions 1–5 assigned; ruling on decision 0001
W3 Testbed (Nov)	CSS / BWS drafting	Harness interface and schema adopted; reference CPU class and block mix; COI policy adopted; node-host call; T1 opened if 0001 passed; 0002 = k + tolerance
W4 Rubric & call text (Dec)	SSR PQC workshop, 13 Dec, Baltimore	Rubric v1; “PQC-ready chain” v1; call-text package tagged and delivered (O5)
W5 Pilot (Jan)	Applications open (planned)	3–5 pilot nodes; tolerance calibrated; rubric v1.1
W6 Retrospective (Feb)	CoDecFin 2027 at FC’27	Phase A retrospective; A2 / B confirmed; BGIN Standard outline v0 (O6); maintainers named

🔗 **Takeaway:** Each session rides an existing calendar anchor. Conveners publish a status on Discourse after each workshop, linking the record.

Integrity — Why Every Action Carries a Signature

Baseline controls

- DCO sign-off on every commit: web setting, client hook, CI job matching sign-off to author
- Cryptographic signatures required on every branch and tag, no bypass; .allowed_signers for offline verification
- main: PR-only, three required checks, linear history, no force-push, no deletion, no admin bypass
- Merged by fast-forward push of the signed head — never the squash or rebase buttons
- Actions SHA-pinned, read-only; Dependabot weekly; OpenSSF Scorecard

What the controls caught

- The first four merges used GitHub's squash button: GitHub-signed commits, three without sign-off — rejected by the history check
- main rebuilt the same hour from the PRs' signed heads: identical tree, verify-history clean
- Approvals 1 $\rightarrow$ 0 while one convener is seated; restored with the second

Every commit and tag on main is signed and signed-off by a named contributor; the whole history verifies offline.

🔗 **Takeaway:** A neutral record is worth citing only if anyone can verify who wrote it.

What Is Held Back, and Why — the Public Rule

The default is publication. Material is held only if at least one of four tests is true; the rule is public even when the material is not. Held material lives in a separate access-controlled repository readable by the conveners and the seated committee.

Test	Meaning	What is published instead
Not ours to publish	Circulated without clearance; Chatham House; third-party rights	The fact and the provenance row
Names a person deliberatively	A nomination, a COI disclosure, a score before announcement	The seated roster row; the fact of a recusal
Would pre-empt an announcement	Draft call text before the host publishes it	The tagged package once acknowledged
Arrived under terms	Applicant material; results tied to a named operator	What the host allows

🔗 **Takeaway:** Unfinished is not confidential. Awkward is not confidential. Disagreement about a metric, a scheme, a date or a ruling is public work.

Outreach and Sources — What Was Said, and Where It Came From

Public channel (since 11 Sep 2026)

- The *Post-Quantum Crypto Agility Competition* category on the BGIN Discourse: launch, workshop series, committee nominations; replies on the IKP PQC and GDC26 threads
- Held for the committee's week: the metric and scheme call for comment; the node-host call
- bgin-global.org/projects/pqc-migration carries the competition: repository, the category, the GDC26 scene-setting deck and the PRX Quantum paper
- House style: no salutation; the post carries the ask and the link, the repository the detail; nothing announced as settled before ratification

Resources with provenance

- METI/NEDO community briefing (July 2026) and its digest
- Google Quantum AI et al., PRX Quantum 7, 031001 — DOI, arXiv, ePrint
- GDC26 meeting report and scene-setting deck (hosted on the BGIN site)
- Project Eleven, state-reuse key recovery (Sep 2026)
- PQC Audit Index: FIPS 206 status, LMS sizes, M-19 checklist seeds, migration timelines
- BIP-340; BIP-360 direction — reference only

🔗 **Takeaway:** Every claim that has a source names it; where sources disagree, both are carried and attributed.

Status at Block 15 — and What Is Open

In the record (main, signed, CI green)

- Charter: README, governance, roster, outcomes, contributing
- Evaluation: register M-01..M-22 with proposed directions, rubric skeleton, COI policy
- Testbed: node model, scheme register, harness spec, result schema, T1 track contract, calibration harness instance
- Roadmap W1–W6; decision 0001 proposed; confidentiality rule; outreach record; resources

Open doors

- **Second convener** — the reason approvals sit at 0; onboarding is docs/SIGNING.md
- **Nominations** — open; seats across academia, Bitcoin, Ethereum, wallets, measurement, governance
- **Node hosts** — the call opens at W3; the roster is empty
- **SSR paper** — 12 Oct deadline; the repository holds a competition-design contribution
- **Decision records** — 0001 at W2; 0002 (*k*, tolerance, reference class) at W3

🔗 **Takeaway:** Twenty-two signed commits, one convener, zero nominations confirmed. The record is ahead of the committee — Block 15 is where that inverts.

How to Contribute — Thread, Issue, Pull Request

1. **Discuss** on the Discourse category; every proposal starts as a thread
2. **Open an issue** from a template: source, metric, scheme, node-host offer, committee nomination
3. **Pull request**, signed and signed-off — one-time setup with scripts/setup-signing.sh; docs/SIGNING.md walks it
4. **Land it** by fast-forward push of the signed head after review; the CHANGELOG records it per signed tag

Repository
github.com/bgin-global/pqc-agility-competition

Discourse category
bgin.discourse.group/c/post-quantum-crypto-agility-competition/44

Documents CC BY 4.0; code Apache-2.0; Contributor Covenant 2.1. Cite a tagged version via CITATION.cff.

🔗 **Takeaway:** Nothing here is final until the evaluation committee ratifies it and the host publishes the official call.

The repository is the record.

Decisions, drafts, sources, and testbed specifications
— signed, public, and draft v0 until ratified.

github.com/bgin-global/pqc-agility-competition · Post-Quantum Crypto Agility Competition category on the BGIN Discourse



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